

PAPER158: Hybrid MUGE Blending Model: $$g_{\text{hybrid}} = g_{\text{compressed}} + (1 - \beta) \cdot g_{\text{resonance}}$$

Session: 47 | Date: March 13, 2026 | Thread: 7f9068 | Domain: §2.3

PAPER158 — Hybrid MUGE Blending Model: $g_{\text{hybrid}} = \beta \cdot g_{\text{compressed}} + (1 - \beta) \cdot g_{\text{resonance}}$

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Abstract

This paper introduces the **Hybrid MUGE Blending Model**, a continuous interpolation between the Compressed MUGE and Resonance MUGE gravity frameworks. The blending parameter $\beta = \exp(-B/B_{\text{crit}})$ transitions smoothly from $\beta \rightarrow 0$ (pure resonance, magnetar regime) to $\beta \rightarrow 1$ (pure compressed Newtonian+, weak-field regime). This unified hybrid is the **first algebraic bridge** between the Newtonian limit and the full resonance superconductive regime within the UQFF framework, extending §2.2 PAPER155 which proved Newtonian emergence only in the $f_{\text{TRZ}} \rightarrow 0$ limit.

1. Motivation

PAPER146 (§2.2) established the 12-term resonance MUGE master equation. PAPER090 (§1.12) established the 9-term compressed MUGE. Previously, no **analytical bridge** existed for intermediate regimes where B is a significant fraction of B_{crit} (e.g., solar magnetosphere, white dwarfs, neutron star crusts).

The blending model provides:

- Continuous function of B/B_{crit} across the full magnetic field parameter space
- Smooth limiting behavior at both extremes
- Single equation for all astrophysical environments

2. Hybrid Blending Equation

$$g_{\text{hybrid}}(r, t) = \beta \cdot g_{\text{comp}}(r, t) + (1 - \beta) \cdot g_{\text{res}}(r, t)$$

where

$$\beta = e^{-B/B_{\text{crit}}}$$

g_{comp} = compressed MUGE gravity (PAPER_090, 9-term Newtonian+corrections)

g_{res} = resonance MUGE gravity (PAPER_146/159, 12/13-term superconductive resonance)

B = local magnetic field strength [T]

B_{crit} = critical quantum field (4.4×10^{13} T for magnetars)

3. Limiting Cases

3.1 Magnetar Regime (B/B_crit → 1)

$\beta = e^{-1} \approx 0.368 \quad \rightarrow \quad g_{\text{hybrid}} \approx 0.37 g_{\text{comp}} + 0.63 g_{\text{res}}$

For SGR 1745-2900 (B = 3×10¹¹ T, B_crit ≈ 4.4×10¹³ T):

$\beta_{\text{SGR}} = e^{-3 \times 10^{11} / 4.4 \times 10^{13}} \approx 0.9933 \approx 1$

→ g_{hybrid} ≈ g_{comp} for SGR 1745 (compressed dominant near-critical but below)

3.2 Solar Regime (B << B_crit)

For Sun (B_s = 10⁻⁴ T):

$\beta_{\text{Sun}} = e^{-10^{-4} / 4.4 \times 10^{13}} \approx 1.000000$

→ g_{hybrid} → g_{comp} (pure Newtonian+ regime)

3.3 Neutron Star Surface (B ~ 10¹² T)

$\beta_{\text{NS}} = e^{-10^{12} / 4.4 \times 10^{13}} \approx 0.977$

→ g_{hybrid} ≈ 0.977 g_{comp} + 0.023 g_{res} (dominantly compressed; 2.3% resonance correction)

4. Python Implementation (CP3)

```
import math
def compute_hybrid_muge(g_compressed: float, g_resonance: float,
B: float, B_crit: float = 4.4e13) -> float:
    """
    Hybrid MUGE blending: g_hybrid = beta * g_comp + (1-beta) * g_res
    where beta = exp(-B/B_crit).
    Args:
    g_compressed: Compressed MUGE gravity [m/s²]
    g_resonance: Resonance MUGE gravity [m/s²]
    B: Local magnetic field [T]
    B_crit: Critical field (default 4.4e13 T, magnetar)
    Returns:
    g_hybrid: Blended MUGE gravity [m/s²]
    """
    if B_crit <= 0:
        raise ValueError("B_crit must be positive")
    beta = math.exp(-B / B_crit)
    return beta * g_compressed + (1.0 - beta) * g_resonance
```

5. Validation — 7-System Blending Table

System	B [T]	β	g_comp [m/s²]	g_res [m/s²]	g_hybrid [m/s²]
SGR 1745-2900	3×10 ¹¹	0.9933	1.783×10 ³	1.655×10 ¹²	1.785×10 ³ +dominated

System	B [T]	β	g_{comp} [m/s ²]	g_{res} [m/s ²]	g_{hybrid} [m/s ²]
Sagittarius A*	1×10^{-8}	~ 1.0	1.816×10^{-3}	1.256×10^{100}	$\approx g_{\text{comp}}$
Tapestry	1×10^{-8}	~ 1.0	2.989×10^{31}	1.257×10^{112}	$\approx g_{\text{comp}}$
Westerlund 2	1×10^{-8}	~ 1.0	2.989×10^{31}	1.257×10^{112}	$\approx g_{\text{comp}}$
Pillars	1×10^{-7}	~ 1.0	1.989×10^2	1.256×10^{10}	$\approx g_{\text{comp}}$
Rings	1×10^{-7}	~ 1.0	2.989×10^{33}	1.257×10^{113}	$\approx g_{\text{comp}}$
Student's Guide	1×10^{-10}	~ 1.0	2.000×10^{11}	1.257×10^{11}	$\approx g_{\text{comp}}$

Note: For all non-magnetar systems, $\beta \approx 1$ so $g_{\text{hybrid}} \approx g_{\text{comp}}$. The blending becomes significant only for systems with $B/B_{\text{crit}} > 0.01$ (i.e., $B > 4.4 \times 10^{11}$ T).

6. Theoretical Significance

The hybrid model provides:

- A unified equation** replacing case-by-case selection of compressed vs. resonance
- Continuous parameter space** enabling interpolation between observational regimes
- Automatic mode selection:** β encodes the magnetic suppression factor from PAPER_090
- **Extends PAPER_155**:** While PAPER_155 proves Newtonian emergence at $f_{\text{TRZ}} \rightarrow 0$, this model handles intermediate B fields without requiring $f_{\text{TRZ}} \rightarrow 0$

Connection to 4 UQFF Operational Modes (PAPER_064):

- Compressed: $\beta = 1$
- Buoyant: β and Ubi terms active
- Resonant: $\beta = 0$
- Superconductive: $\beta = 0$, f_{TRZ} active

Status: ■ Complete | **CP Stage:** CP3 (new `compute_hybrid_muge()` function) **Supersedes:** N/A (new model) | **Related:** PAPER090 (compressed), PAPER146 (12-term resonance), PAPER155 (Newtonian limit), PAPER064 (4 modes)

UQFF computed: MUGE buoyancy ratio $Ubi/FU = [SSq]^{1/2} r / GM = 5.7e-1 \times 5.0e-4 = 2.85e-4$; compressed MUGE baseline $g = 5.4e-7$ m/s² at r_{ISCO} .